Table 4 — Benchmarking Methods and the Repository

BENQODHI Day 2 morning round tables · 11 September 2026

Printable guide for the table. Keep it in front of you during the discussion.

Aim of the round tables

Collect focused expert input for a possible BENQODHI output (paper, report, roadmap or policy note). The central question is how important challenges in health and the life sciences can become **credible, reusable benchmarks** for classical, AI-based, hybrid and quantum algorithms.

We are **not** building the benchmark repository during the workshop. The goal is to identify the methods, measures and infrastructure choices that would make a future benchmark repository useful and credible.

This table's role

Table 4 gathers people experienced in benchmarking computational methods — standard classical, AI-based, hybrid and quantum. Its role is to define how these approaches should be compared on candidate problems, and to identify a practical route for getting BENQODHI benchmark material online and keeping it credible over time.

Questions to answer

1. **Comparing approaches** — How should standard classical, AI-based, hybrid and quantum approaches be compared on the same problems?
2. **What to measure** — Which measures matter most: solution quality, running time, efficiency, energy consumption, resources used, reproducibility?
3. **Methods to include** — From your point of view, which best current methods and new approaches should be included?
4. **Online and credible** — What is the simplest route to put benchmark material online, and who should maintain or review it?

Transversal question — answer once for the table

Sustaining the ecosystem (infrastructure, community, scientific governance): BENQODHI only works if what this table produces feeds one connected, durable ecosystem rather than a silo. Of the three pillars — the shared **infrastructure** that hosts and maintains it, the **community** that keeps contributing, and the **scientific governance** that keeps it credible — how can this table best feed the third pillar, scientific governance? What would it take for the problems, data and baselines from this table to be curated, reviewed and trusted as credible science over time, and what governance role should this table's own community play in setting and upholding those standards?

Expected output

- short answers to the questions above;
- a list of measures that should be used to compare approaches;
- a list of current or new methods that should be included;
- a practical route to put the benchmark material online and keep it credible.

A worked example (`answers.md`) lives in this folder to show the level of detail expected. Note takers: draft answers live into `answers.md`. Rapporteur: report the main conclusion, main infrastructure need, why it matters, the most important missing piece, the next action, and the governance answer.